

RESEARCH PRESENTATION

Lightweight Spatial-Temporal Fused Transformer for Urban Pedestrian Flow Anomaly Detection

A parameter-efficient STF architecture adapted to small-sample spatiotemporal forecasting on the BJ-Taxi dataset

OUTLINE

Agenda

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Urban pedestrian flow anomaly detection problem

02 Model Family Comparison

Statistical vs. Neural vs. Deep learning approaches

03 Lightweight STF Architecture

Cross-attention fused spatial-temporal token design

04 Anomaly Detection Module

Residual + reconstruction fusion at 9:1

05 Deployment & Toolchain

FastAPI backend, Streamlit frontend, Optuna, SHAP

06 Summary & Next Steps

Key contributions and future directions

Detecting anomalous pedestrian flow in urban networks

Problem

Urban pedestrian flow exhibits strong spatial-temporal dependencies. Sudden deviations — rallies, accidents, concerts — are critical for city management but rare and hard to forecast with standard methods.

Dataset & Pipeline

Built on the BJ-Taxi (TaxiBJ P4) dataset with 1024 grid nodes. The pipeline runs data cleaning and feature engineering first, then feeds the cleaned tensor into a forecasting backbone for downstream anomaly scoring.

Key challenge: the BJ-Taxi dataset has a limited sample size. Large pre-trained spatiotemporal transformers easily overfit, while simpler models fail to capture long-range spatial-temporal patterns.

Three generations of spatiotemporal models

Family	Strengths	Limitations
Statistical (ARIMA, SVR)	Simple, interpretable, fast to train on small data	Cannot predict emergent events; weak spatial interaction modeling
Neural Network (GCN, GRU, LSTM)	Captures local spatial correlations well	Limited long-range temporal dependence; static graph discards dynamic info
Deep Learning (Transformer, AGFormer)	Multi-modal fusion; long-horizon forecasting; flexible spatial-temporal coupling	Large parameter count; overfits on small datasets like BJ-Taxi

Direction: fuse deep-learning expressiveness with parameter efficiency for small-data settings.

AGFormer is strong, but too heavy for BJ-Taxi

Baseline: AGFormer

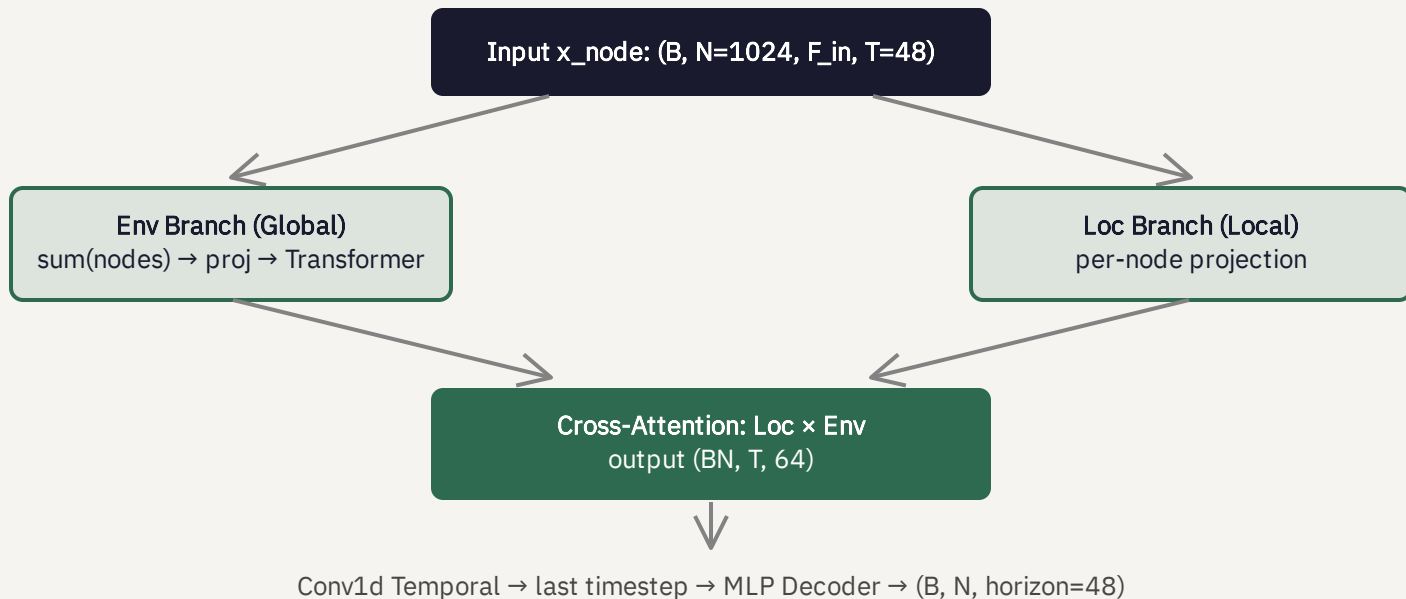
- Dynamic-graph attention with parallel Transformer stacks
- Spatial-temporal token fusion + self-attention cross-correlation
- Augmented with static graph / GCN modules
- Large parameter footprint

Problem on BJ-Taxi

- Limited sample volume in the BS-Txi / BJ-Taxi split
- Fine-tuning a large model easily overfits
- Many parameters are redundant for this scale of graph
- Need a lightweight alternative with comparable accuracy

Our approach: Lightweight STF — Spatial-Temporal Factorized Transformer — achieves better prediction than AGFormer with significantly fewer parameters, adapted specifically to the BJ-Taxi data scale.

Dual-branch factorization: Env global + Loc local



Design note: instead of stacking full self-attention over all nodes, STF factorizes global context (Env) from local node features (Loc) and fuses them with one cross-attention layer.

Tensor shape flow through STF

Stage	Operation	Tensor Shape
Input	Raw node features	$(B, 1024, F_{in}, 48)$
Env Branch	sum over nodes, proj, Transformer encoder	$(B, T, 64)$
Loc Branch	per-node linear projection	$(BN, T, 64)$
Cross-Attention	Loc queries attend to Env context	$(BN, T, 64)$
Conv1d + Pool	Temporal conv, take last timestep	$(B, N, 64)$
MLP Decoder	Per-node horizon regression	$(B, N, 48)$

Three anomaly scores fused at 9:1

Statistical Threshold

Time-segmented z-score over historical windows. Flags local outliers within each hour / day-part.

Prediction Residual

|actual - STF forecast|. Focuses on global, sustained, large-scale anomalies; local spikes contribute little.

Reconstruction Error

Autoencoder reconstruction residual. Complements residual by capturing atypical manifold structure.

Fusion: weighted score = $0.9 \times \text{Residual} + 0.1 \times \text{Reconstruction}$

Known limitation: anomalies are evaluated on manually injected spikes. The baseline accuracy is sub-optimal because multiple control groups are insufficient — future work will add more anomaly types and comparison baselines.

Serving stack and analysis tools

Serving Architecture

- **FastAPI** — backend inference service, exposes forecast and anomaly-score endpoints
- **Streamlit** — frontend visualization dashboard for real-time flow map and anomaly alerts

Analysis & Optimization Tools

- **Optuna** — hyperparameter search for learning rate, dropout, attention dims
- **SHAP** — model interpretability, feature-level anomaly attribution
- **Attention hooks** — extract average attention distribution across heads on the validation set

The attention-hook analysis reveals which spatial regions the model relies on during anomaly scoring, complementing SHAP's feature-level explanations and providing a qualitative check on whether the fused Env-Loc attention is meaningful.

Key contributions and next steps

01 **Lightweight STF beats AGFormer on BJ-Taxi with fewer parameters**

Dual-branch Env/Loc factorization + single cross-attention avoids full pairwise self-attention cost.

02 **Fused anomaly scoring combines prediction residual and reconstruction**

Weighted 9:1 fusion captures both sustained large-scale anomalies and atypical manifold deviations.

03 **End-to-end pipeline from data to deployment**

FastAPI + Streamlit serving, Optuna tuning, SHAP + attention-hook interpretability.

Next steps: expand anomaly injection types and control groups, validate on additional city datasets, and experiment with dynamic graph updates in the Env branch.

Thank You

Questions & discussion welcome